

Real-World Clinical and Economic Impact of Oral Antidiabetic Drug Optimization for Insulin De-intensification in Type 2 Diabetes: A Retrospective Cohort Study in Malaysian Public Hospital

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TAKE-HOME MESSAGE

OGLD optimisation improved glycaemic control while reducing insulin exposure.

0.70% vs 0.16%

mean HbA1c reduction
(absolute % points)
OOG vs SCG | p=0.036

−10.3 vs +4.2

mean change in
insulin dose (U/day)
OOG vs SCG | p<0.001

RM75.78

incremental cost per
1% HbA1c reduction
(direct drug costs)

INTRODUCTION

23.1% → 30.3%

insulin use in Malaysian public primary care, 2013–2019²

Guidelines advise optimising oral glucose-lowering drugs (OGLDs) before escalating insulin; clinical inertia often prevents this.¹

OBJECTIVE

Evaluate the feasibility, clinical impact and direct-drug cost-effectiveness of structured OGLD optimisation.

METHODOLOGY

- Single-centre retrospective cohort, Hospital Taiping; Sep–Dec 2024, 6-month follow-up
- 216 adults with type 2 diabetes mellitus (T2DM) on insulin ≥1 year
- OGLD-optimised (OOG) n=107 vs standard care (SCG) n=109; baseline eGFR, lipids and prior OGLD use comparable

STEPWISE PROTOCOL

Pharmacist-screened; final decision with the attending clinician.

- 1

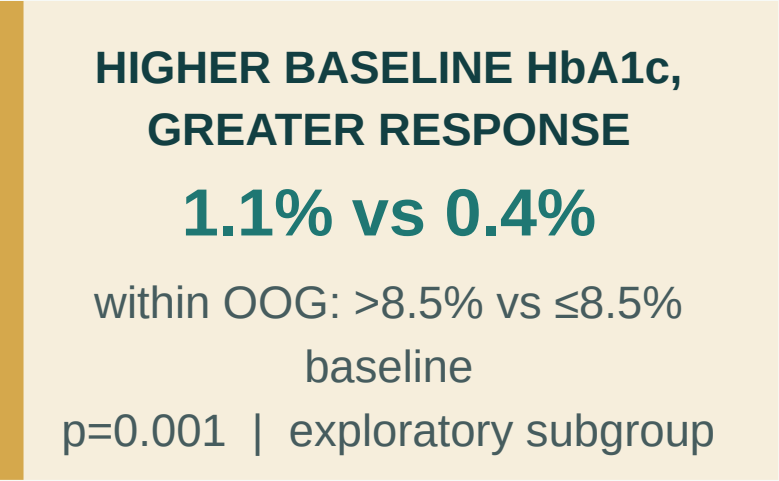
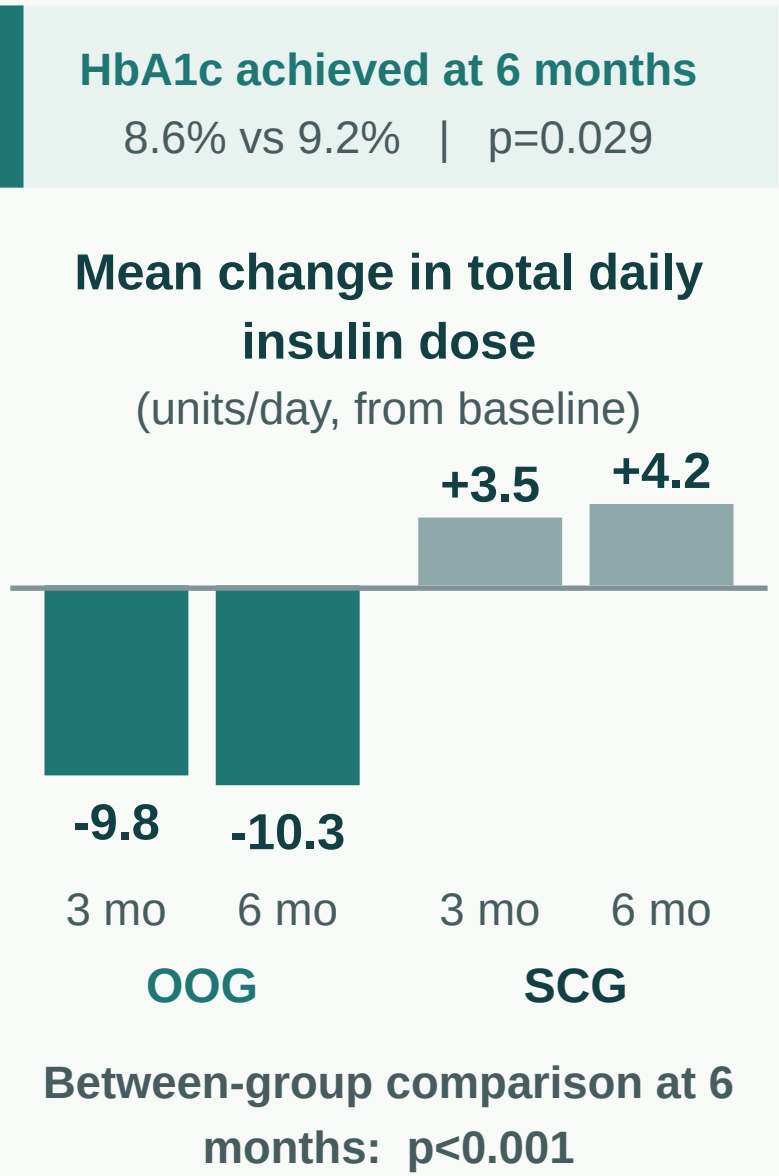
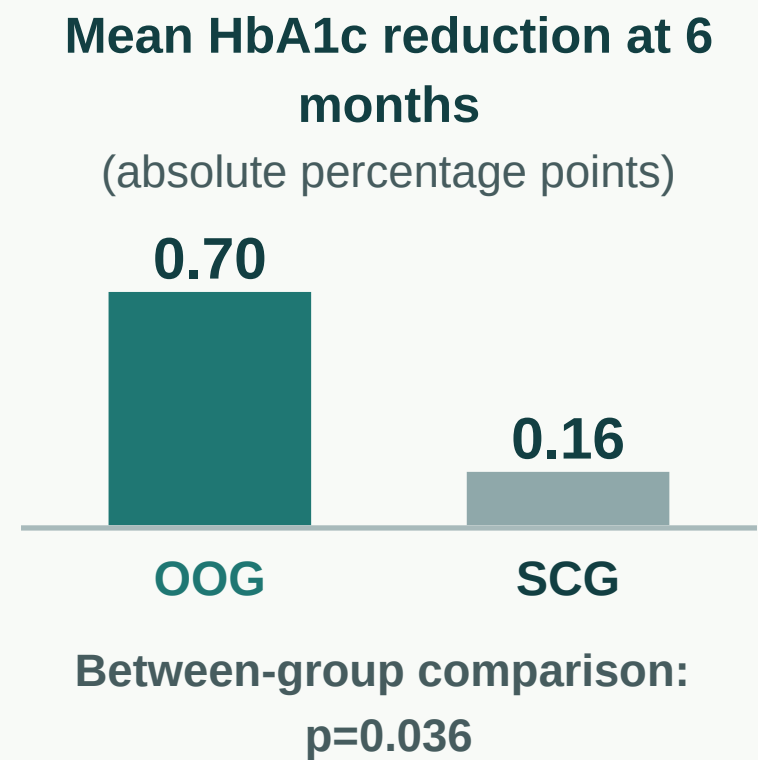
MAXIMISE OGLDs Titrate to maximally tolerated doses.
- +
- 2

ADD SGLT2i OR DPP4i If not already prescribed.
- ↓
- 3

DE-INTENSIFY BY TDD Total daily dose (TDD) ≤10 U/day: prioritise discontinuation. TDD >10–20 U/day: reduce basal by 50%.

RESULTS

In the OOG (n=107), optimisation most often meant adding an SGLT2 inhibitor (85.0%, n=91) or a DPP4 inhibitor (11.2%, n=12).

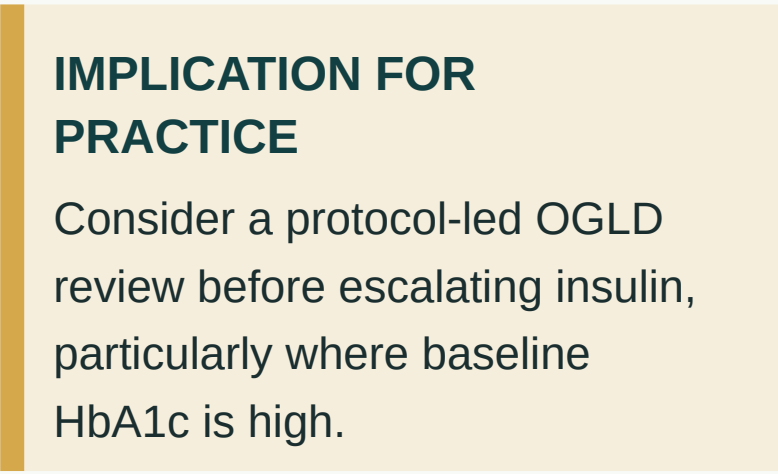
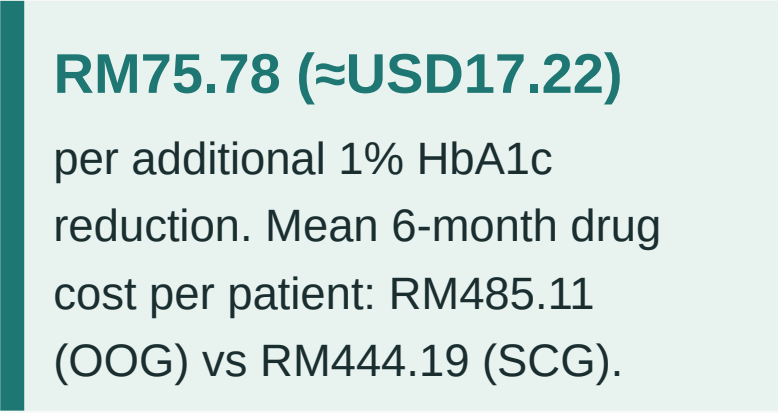
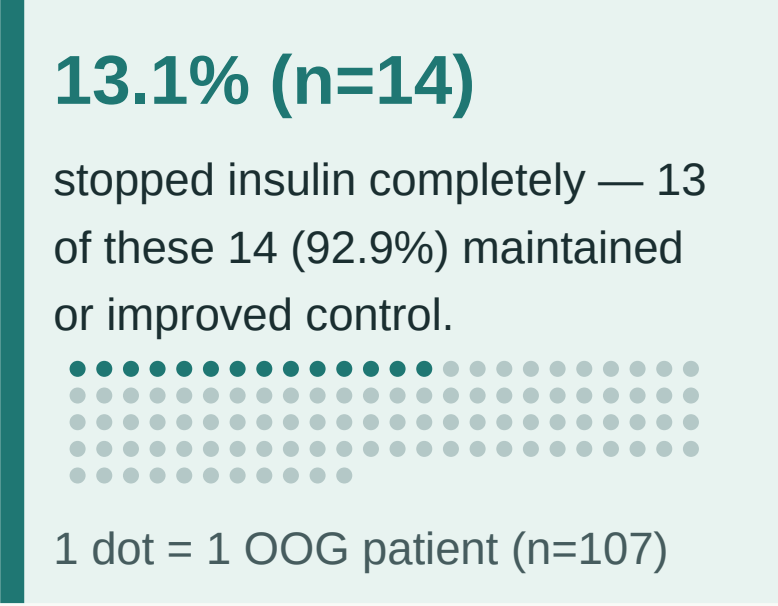


CONCLUSION

Compared with standard care, structured OGLD optimisation was associated with a larger HbA1c fall and a clinically meaningful drop in insulin exposure.³

The association was strongest at baseline HbA1c >8.5% (exploratory subgroup).

Costs reflect direct drug acquisition only, consistent with Malaysian cost-effectiveness data for SGLT2i.⁴



LIMITATIONS

Single-centre; retrospective, non-randomised allocation (selection bias); 6-month follow-up; hypoglycaemia not systematically quantified; direct drug costs only.

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